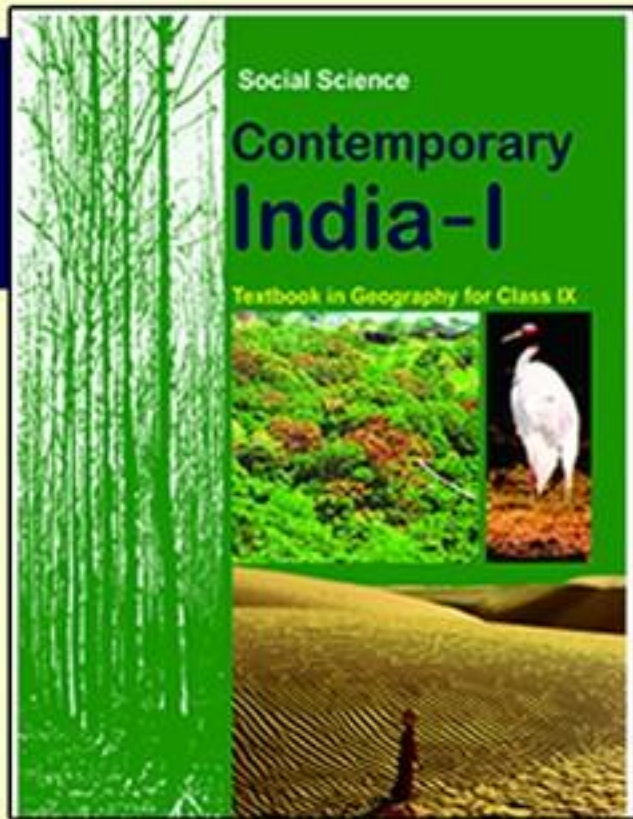




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NCERT BASICS



GEOGRAPHY

SOILS: LAYERS, TYPES AND
VARIOUS CROPS IN SOILS

MODULE - 6





SOIL ... THE SKIN OF THE EARTH!

WHAT IS SOIL?

- Soil is a mixture of minerals, dead and living organisms (organic materials), air, and water.
- It consists of inorganic particles and organic matter.
- Soil is at the interface between the atmosphere, lithosphere, hydrosphere, and biosphere.



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SOIL ... INTERFACE BETWEEN THE MAJOR DOMAINS OF THE EARTH!

HOW THE SOIL IS FORMED?

- Soil minerals form the basis of soil.
- Soil formation or pedogenesis depends first on weathering.
- Weathering is the breaking down of rocks and minerals on Earth's surface.



FACTORS RESPONSIBLE FOR FORMATION OF SOIL

Parent Rock:

Determines colour, texture, chemical properties, mineral content, permeability.

Flora, Fauna and Micro-organism:

Affect the rate of humus formation.

Climate:

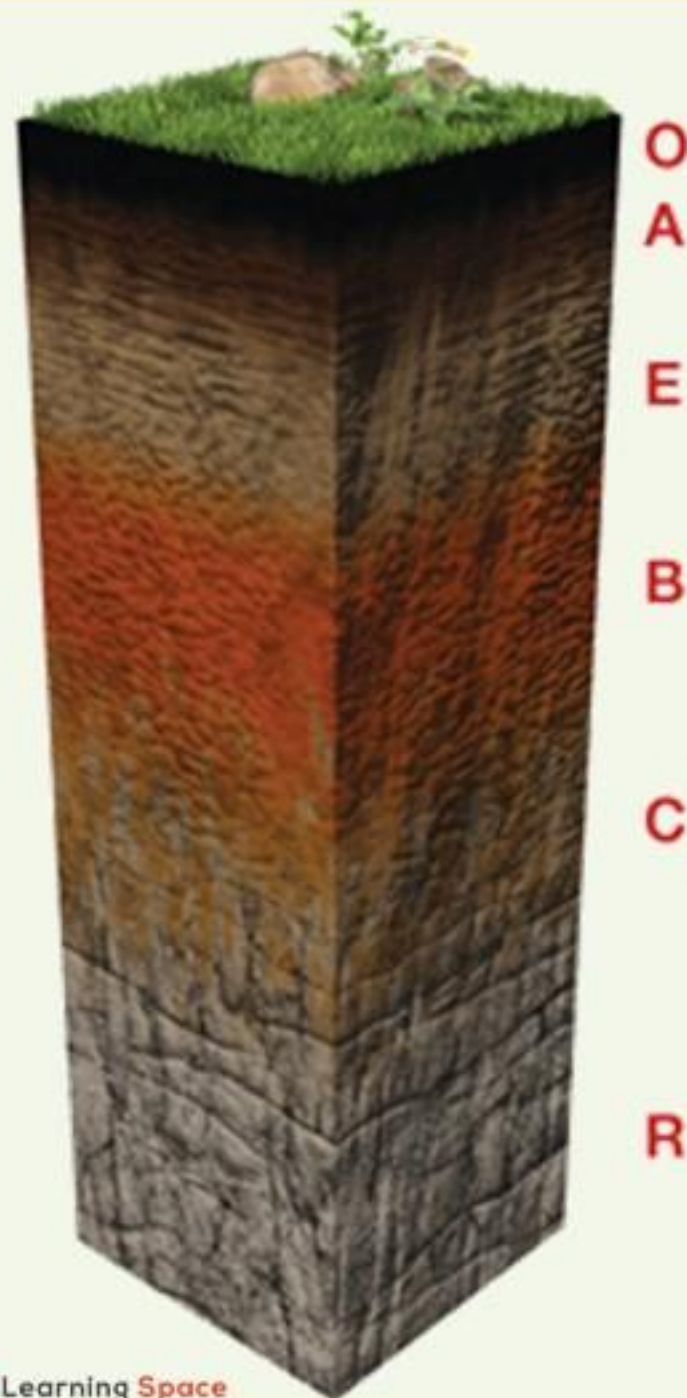
Temperature and rainfall influence rate of weathering and humus formation.

Relief:

Altitude and slope determines accumulation of soil.

Time:

Determines thickness of soil profile.



NOW LET US UNDERSTAND THE LAYERS OF THE SOIL AND ITS IMPORTANCE!

- ✓ The soil is divided into various different layers top to bottom and this arrangement is termed as the **soil horizons**.
- ✓ These layers or horizons are represented by letters O, A, E, B, C and R.



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SOIL PROFILE

THE O-HORIZON:

- It is the superficial or topmost layer of the soil.
- It is primarily composed of decaying organic matter, living organism, and fresh soil.
- This layer hosts most of the roots of small vegetation cover like grass.
- It contains about 20% of organic matter.

THE A-HORIZON:

- This is the second layer of the soil.
- It is composed of highly decaying organic matter and minerals.
- It is the layer that hosts many plants and supports many agricultural and grassland areas.



O (humus or organic)
A (topsoil)

E (eluviated horizon)

B (subsoil)

C (parent material)

R (bedrock)





SOIL PROFILE

THE E-HORIZON:

- ✓ This is the third and the thinnest soil layer.
- ✓ It is termed as the **eluviation** layer due to the leaching of minerals.
- ✓ It is mainly made up of silt and sand after losing most of its minerals as they drip down through the soil.

THE B-HORIZON

- ✓ It is the fourth layer and is also referred to as the sub-soil.
- ✓ It is mainly made up of large rocks, clay, and bedrock.
- ✓ The sub-soil accumulates a lot of clay, aluminum, iron, and organic compounds from the layers above it.
- ✓ It happens from the dripping down of mineralized water through a process known as **illuviation**.



O (humus or organic)
A (topsoil)

E (eluviated horizon)

B (subsoil)

C (parent material)

R (bedrock)





SOIL PROFILE



O (humus or organic)
A (topsoil)

E (eluviated horizon)

B (subsoil)

C (parent material)

R (bedrock)

THE C-HORIZON

- This is the fifth layer of the soil and it also known as the regolith.
- Plant roots do not go through this layer.
- It is made up of fragmented pieces of bedrock.
- Also, it has compacted sediment and cemented geological material.

THE R-HORIZON

- It is the sixth and final layer of the soil.
- It is also known as the bedrock
- This layer is beyond 50 meters from the top-soil.
- Some of the rocks found here include granite, limestone, and basalt.



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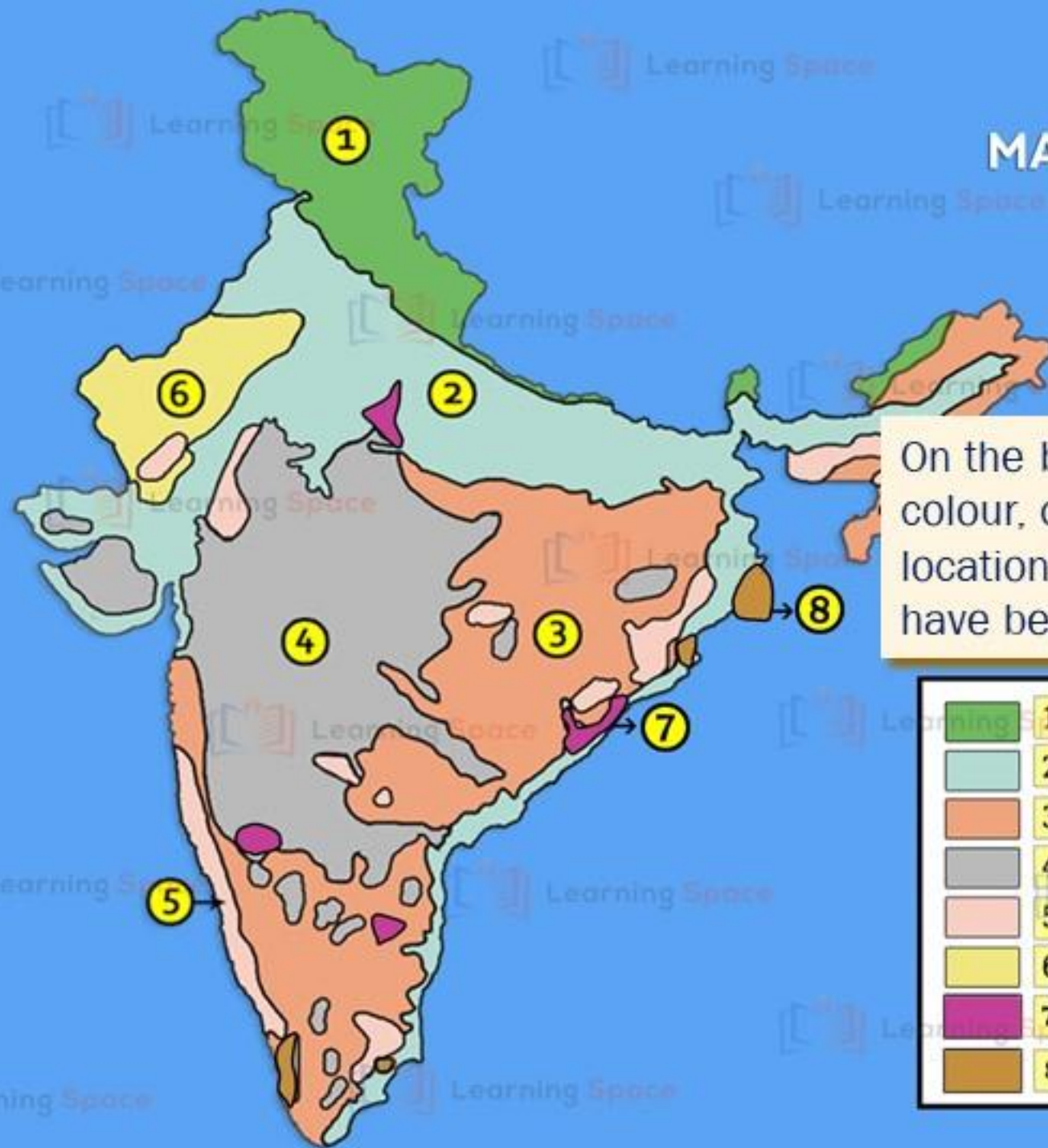


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INDIA

MAJOR SOIL TYPES

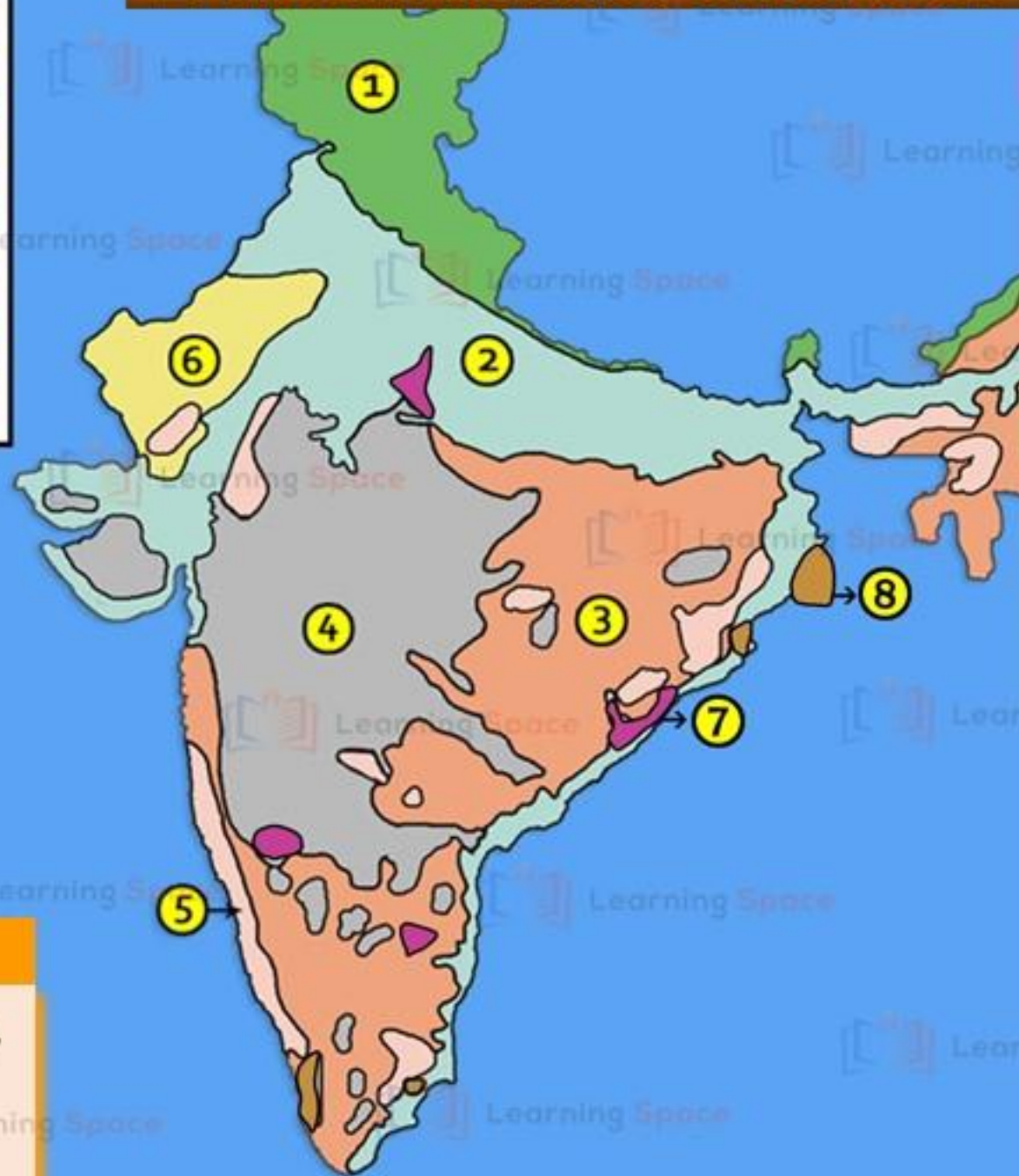


On the basis of genesis, colour, composition and location, the soils of India have been classified into:

1. Forest and Mountain
2. Alluvial Soil
3. Red and Yellow Soil
4. Black Soil
5. Laterite Soil
6. Arid and Desert
7. Saline Soil
8. Peaty Soil

1. MOUNTAIN AND FOREST SOIL

- | |
|------------------------|
| 1. Forest and Mountain |
| 2. Alluvial Soil |
| 3. Red and Yellow Soil |
| 4. Black Soil |
| 5. Laterite Soil |
| 6. Arid and Desert |
| 7. Saline Soil |
| 8. Peaty Soil |



FOUND IN

- Himalayan area
- Western and Eastern Ghats
- A few regions of the Peninsular Plateau.

CHARACTERISTICS

- Acidic soil.
- Rich in organic matter.
- Poor in lime, phosphorus and potash.

SUITABLE CROPS

Wheat, barley, maize, tea, coffee, spices, tropical and temperate fruits.

2. ALLUVIAL SOIL

CHARACTERISTICS

- ✓ Mix of sandy loam and clay soil.
- ✓ Rich in humus (organic matter) and phosphoric acid.
- ✓ Poor in potash and nitrogen.

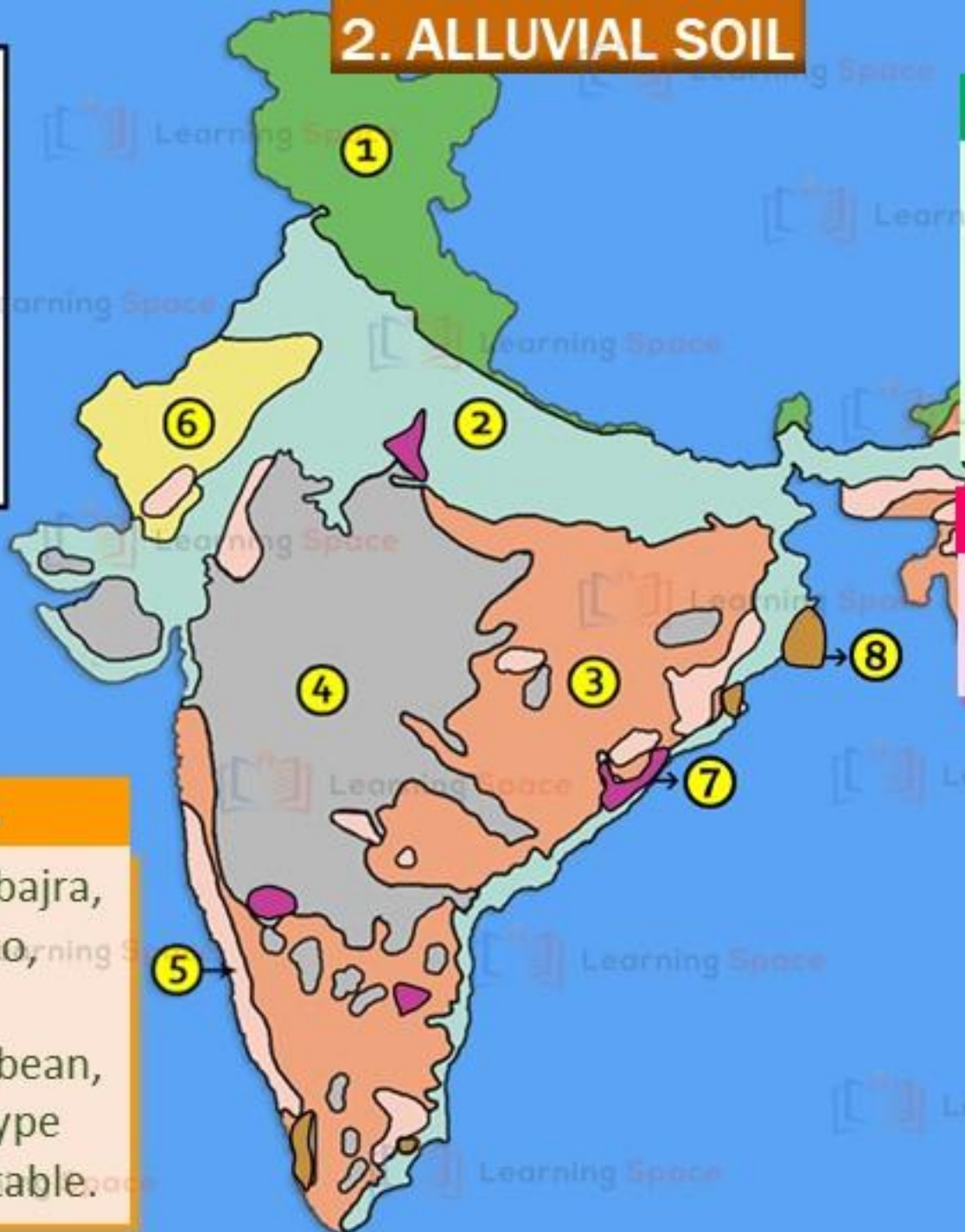
FOUND IN

Most of the delta areas of North India.

SUITABLE CROPS

Cotton, wheat, sorghum, bajra, maize, barley, jute, tobacco, green and black gram, chickpea, pigeon pea, soybean, sesame, groundnut, any type of oilseed, fruit, and vegetable.

	1. Forest and Mountain
	2. Alluvial Soil
	3. Red and Yellow Soil
	4. Black Soil
	5. Laterite Soil
	6. Arid and Desert
	7. Saline Soil
	8. Peaty Soil



3. RED AND YELLOW SOIL

- | | |
|--|------------------------|
|  | 1. Forest and Mountain |
|  | 2. Alluvial Soil |
|  | 3. Red and Yellow Soil |
|  | 4. Black Soil |
|  | 5. Laterite Soil |
|  | 6. Arid and Desert |
|  | 7. Saline Soil |
|  | 8. Peaty Soil |

SUITABLE CROPS

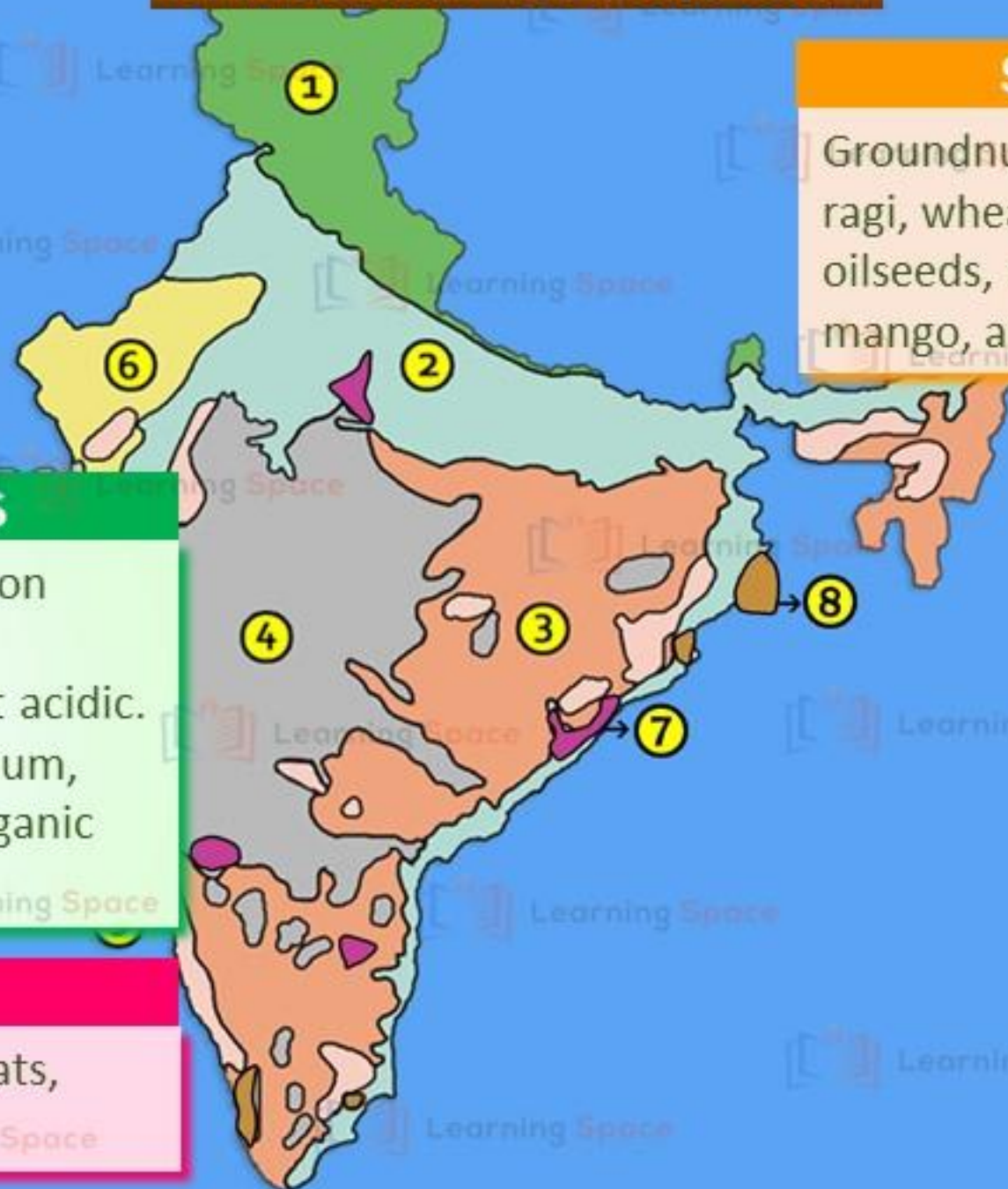
Groundnut, potato, maize/corn, rice, ragi, wheat, millets, pulses, sugarcane, oilseeds, and fruits like citrus, orange, mango, and vegetables.

CHARACTERISTICS

- The soils are red due to iron oxide in them.
- Rich in potash. Somewhat acidic.
- Poor in nitrogen, magnesium, lime, phosphorus, and organic matter.

FOUND IN

Deccan plateau, Western Ghats, Orissa, Chhattisgarh.



4. BLACK SOIL

- | | |
|--|------------------------|
|  | 1. Forest and Mountain |
|  | 2. Alluvial Soil |
|  | 3. Red and Yellow Soil |
|  | 4. Black Soil |
|  | 5. Laterite Soil |
|  | 6. Arid and Desert |
|  | 7. Saline Soil |
|  | 8. Peaty Soil |

CHARACTERISTICS

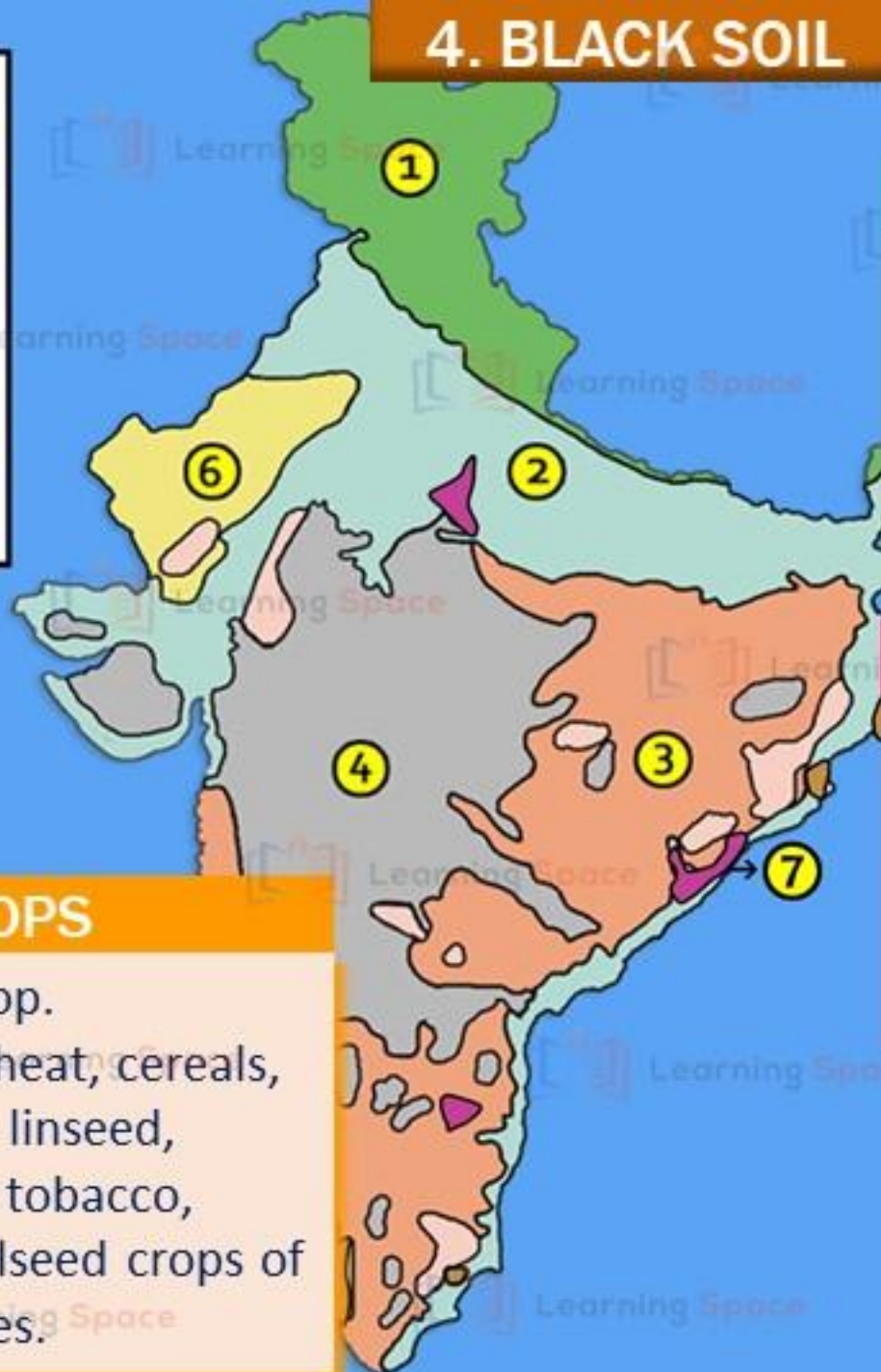
- These soils are formed when lava rocks weather away.
- Rich in magnesium, iron, aluminum, and lime. Poor in nitrogen, phosphorus, and organic matter

FOUND IN

- Deccan lava tract.
- This includes states of Gujarat, Madhya Pradesh, Maharashtra, Andhra Pradesh, Tamil Nadu, and Telangana.

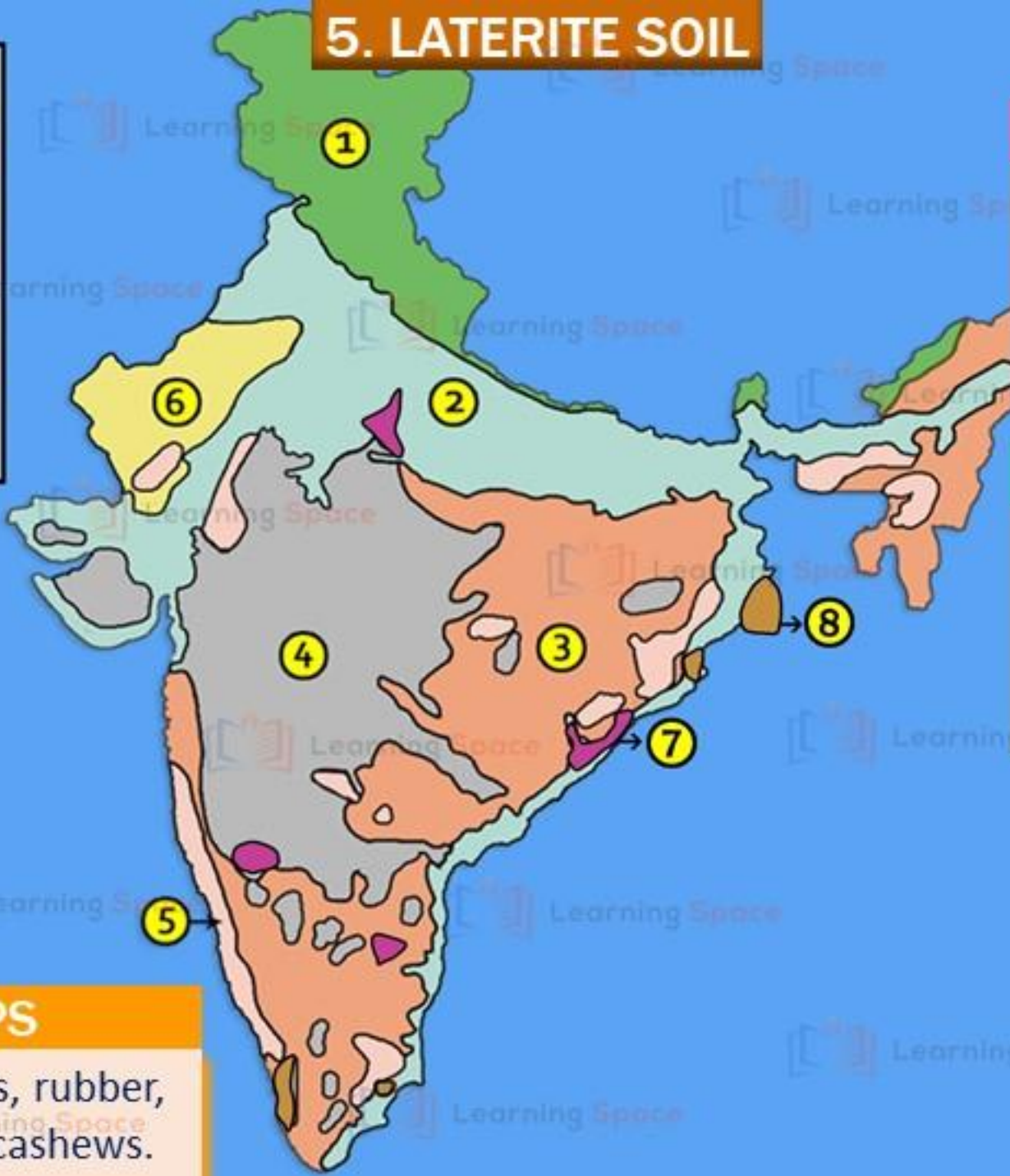
SUITABLE CROPS

- Cotton is the major crop.
- Other crops include wheat, cereals, rice, jowar, sugarcane, linseed, sunflower, groundnut, tobacco, millets, citrus fruits, oilseed crops of all kinds, and vegetables.



5. LATERITE SOIL

	1. Forest and Mountain
	2. Alluvial Soil
	3. Red and Yellow Soil
	4. Black Soil
	5. Laterite Soil
	6. Arid and Desert
	7. Saline Soil
	8. Peaty Soil



FOUND IN

In the states of Madhya Pradesh, Kerala, Karnataka, Tamil Nadu, Assam, and Orissa.

CHARACTERISTICS

- ✓ Acidic soils, rich in iron.
- ✓ Poor in organic matter, calcium, nitrogen, and phosphate.

SUITABLE CROPS

Cotton, wheat, rice, pulses, rubber, tea, coffee, coconut, and cashews.

6. ARID SOIL

SUITABLE CROPS

Barley, maize,
wheat, millets,
cotton, and pulses.

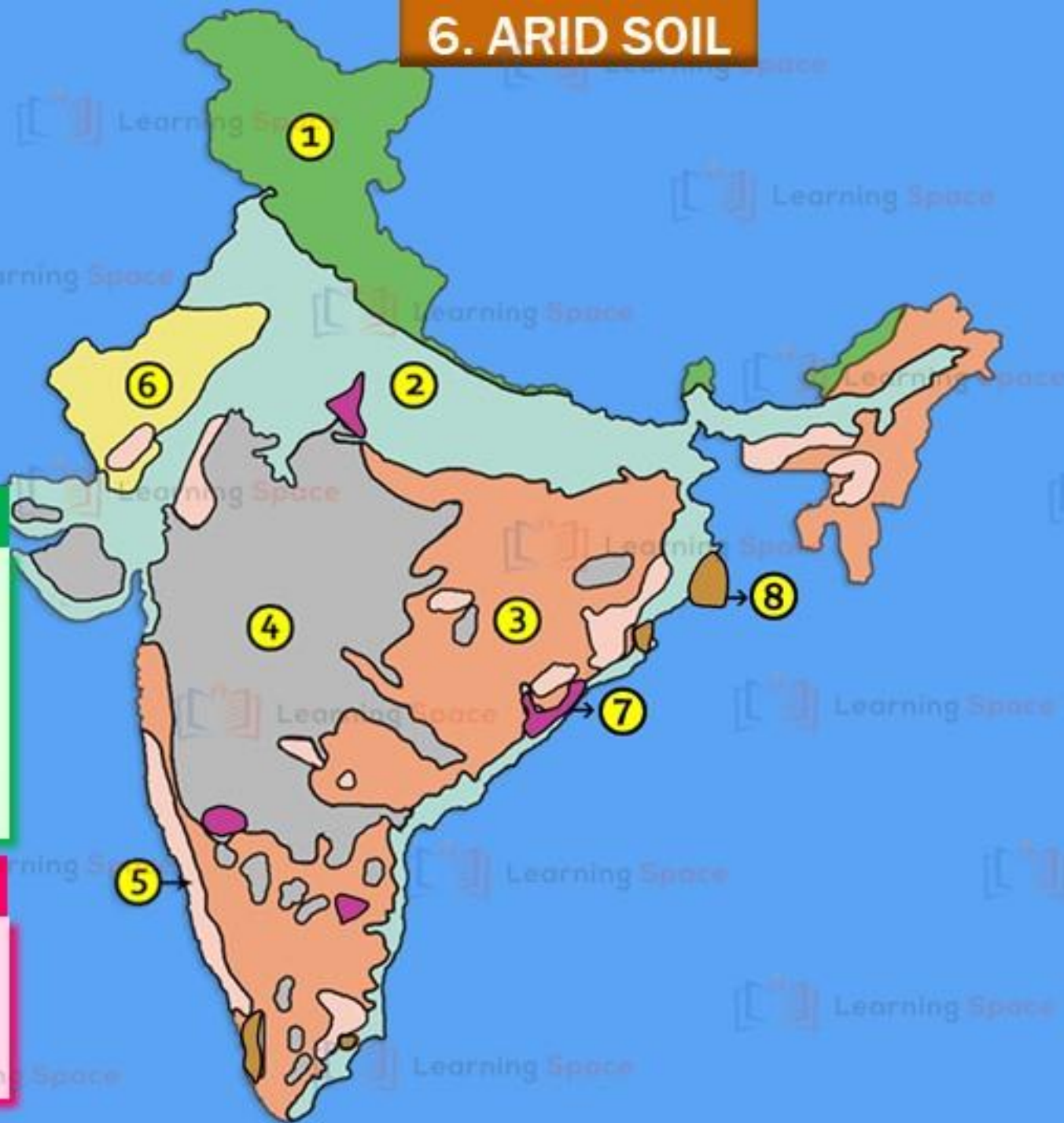
- | |
|------------------------|
| 1. Forest and Mountain |
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| 7. Saline Soil |
| 8. Peaty Soil |

CHARACTERISTICS

- Dry and sandy soil.
- Nitrogen is insufficient and the phosphate
- Rich in plant food and high salt.

FOUND IN

Western Rajasthan, which exhibit characteristic arid topography.



7. SALINE SOIL

FOUND IN

- ✓ Occurs in the State of Uttar Pradesh.
- ✓ Coastal Saline soils occur in the States of Gujarat and West Bengal.

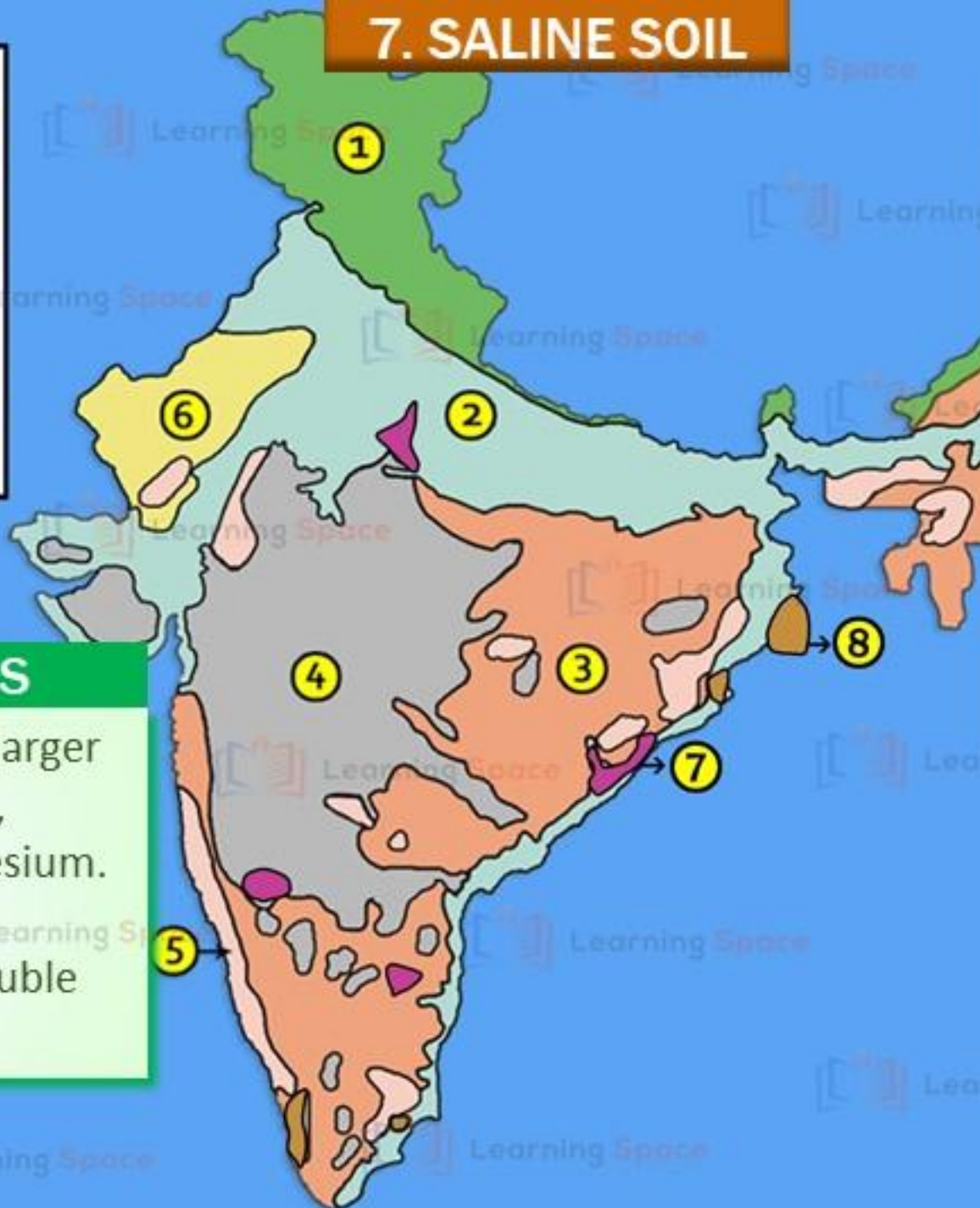
SUITABLE CROPS

Barley, sugarcane, sugar beet, oats, sesbania and other leguminous crops.

CHARACTERISTICS

- ✓ Saline soils contain a larger proportion of sodium, potassium and magnesium.
- ✓ They are infertile.
- ✓ Excessive levels of soluble salts in the soil water.

	1. Forest and Mountain
	2. Alluvial Soil
	3. Red and Yellow Soil
	4. Black Soil
	5. Laterite Soil
	6. Arid and Desert
	7. Saline Soil
	8. Peaty Soil



8. PEATY SOIL

FOUND IN

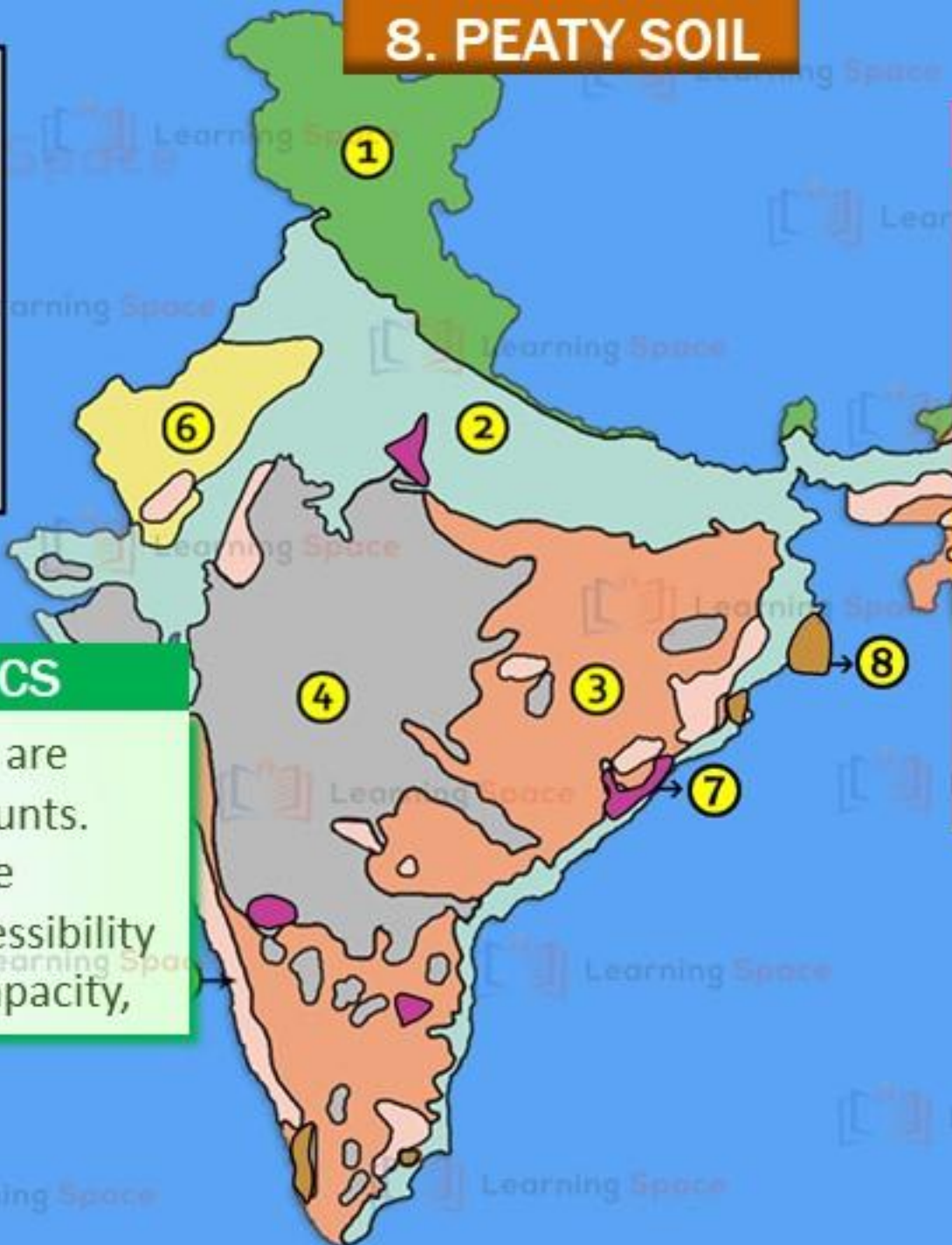
- Northern part of Bihar
- Southern part of Uttaranchal
- The coastal areas of West Bengal
- Orissa and Tamil Nadu.

SUITABLE CROPS

Potatoes, sugar beet, celery, onions, carrots, lettuce and market garden crops are commonly grown

CHARACTERISTICS

- Carbon and nitrogen are present in large amounts.
- High natural moisture content, high compressibility and water-holding capacity,

- 
- A map of India with different regions color-coded to represent various soil types. The legend on the left lists eight soil types: 1. Forest and Mountain (green), 2. Alluvial Soil (light blue), 3. Red and Yellow Soil (orange), 4. Black Soil (grey), 5. Laterite Soil (pink), 6. Arid and Desert (yellow), 7. Saline Soil (purple), and 8. Peaty Soil (brown). The map shows these distributions across the country, with Peaty Soil (8) concentrated in the northern part of Bihar, southern part of Uttaranchal, and coastal areas of West Bengal, Orissa, and Tamil Nadu.
1. Forest and Mountain
 2. Alluvial Soil
 3. Red and Yellow Soil
 4. Black Soil
 5. Laterite Soil
 6. Arid and Desert
 7. Saline Soil
 8. Peaty Soil



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